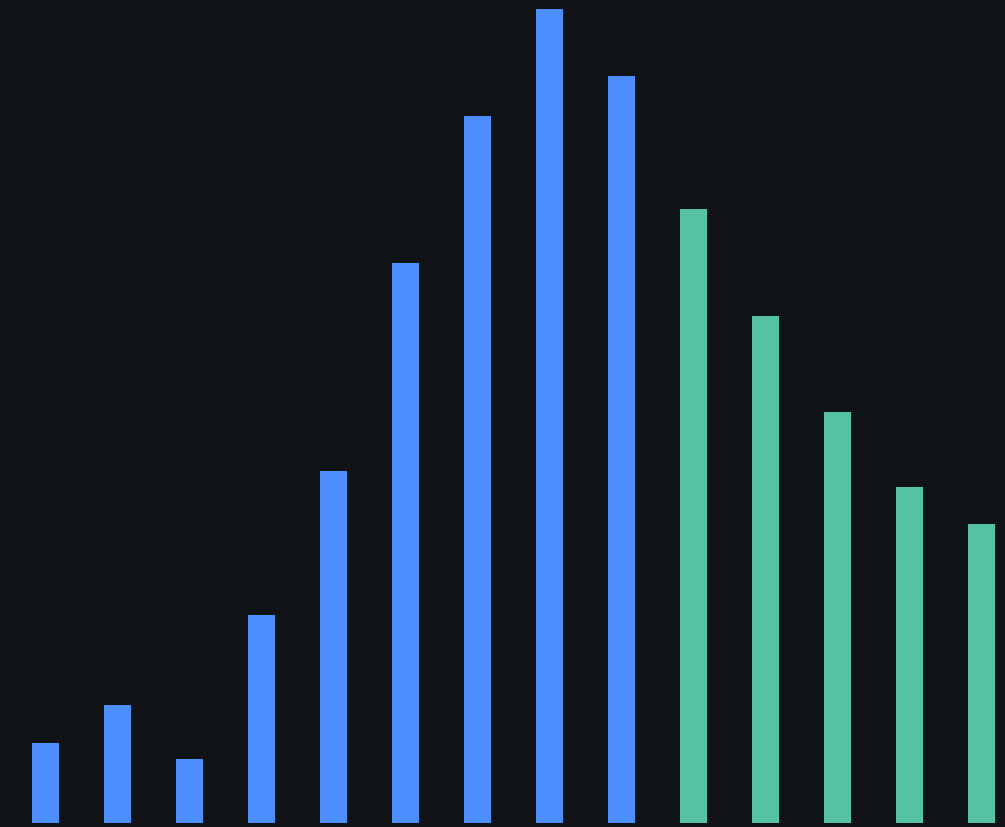


See the effort. Understand the recovery.

A live view of pulse, movement and workload.
Built around a Raspberry Pi.



EFFORT

RECOVERY

Concept graphic - not recorded data

The number is only part of the story.

PULSE

How hard?

Read cardiovascular response.

MOVEMENT

What changed?

Track the activity context.

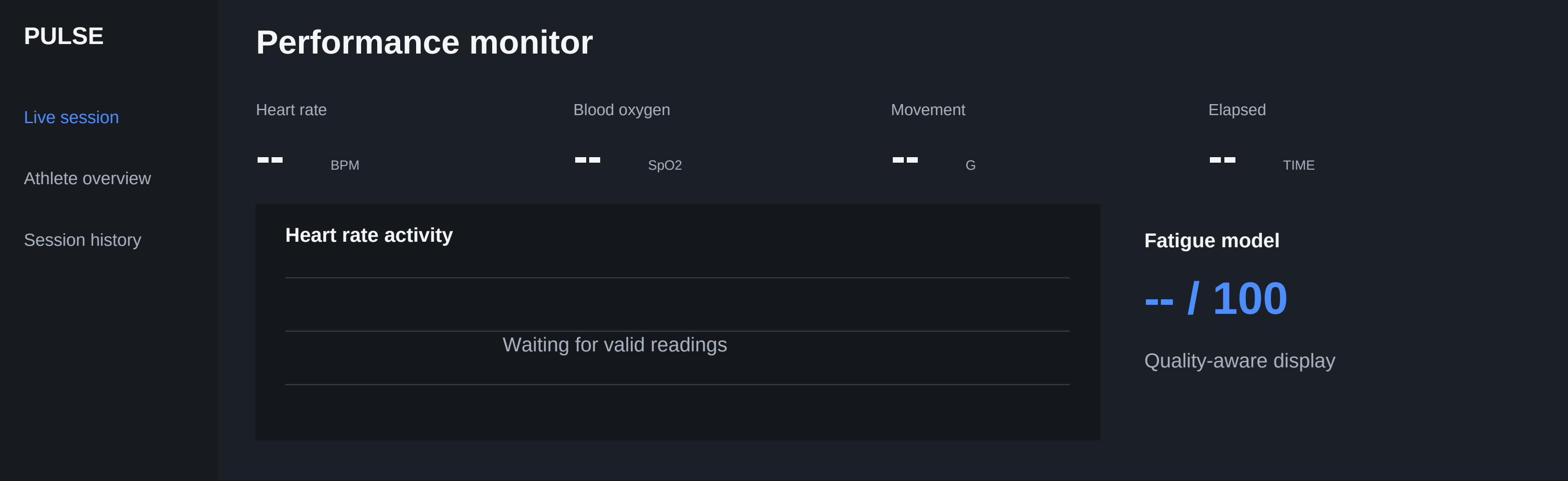
RECOVERY

What next?

Observe after stopping.

PULSE brings these observations into one session a coach can review.

One session. A shared view.



Dashboard schematic based on the implemented interface; no sample readings shown.

LIVE

Watch signals together.

REVIEW

Switch athletes and sessions.

EXPORT

Take the evidence away.

From wearable to browser. Inference stays on the Pi.

01

SENSE

MAX30100 optical sensor
ADXL345 accelerometer

02

> **TRANSPORT**

C++ acquisition
UDP + shared memory

03

> **INTERPRET**

Python signal checks
Random Forest model

04

> **DISPLAY**

Node.js + WebSockets
Local browser dashboard

One pipeline for the real sensors and the clearly labeled stress simulator.

Current web dependencies: fonts and PDF library load externally. Full offline packaging still needs testing.

Rest. Move. Recover.

01

REST

Capture a stable baseline and a simultaneous reference pulse.

02

EXERCISE

Mark the activity. Watch pulse, movement and signal quality.

03

RECOVERY

Mark the stop. Follow the change from recovery onset.

[End session](#)[Show history](#)[Export evidence](#)

An alert depends on the observed signals. A short trial may produce no alert.

Give the alert context.

The model produces a score.
The interface adds observable trends.

Enough valid rest samples are required
before baseline comparisons appear.

Pulse versus rest

Change from the recorded resting average.

Recent movement trend

Change in acceleration magnitude, including gravity.

Recovery response

Pulse change since recovery was marked.

These descriptions accompany the alert. They are not model feature attribution or evidence of causation.

A missing reading is an honest answer.

VALID CONTACT

Show accepted
measurements.

REMOVE CONTACT

Poor signal. Values clear.

RESTORE CONTACT

Wait for valid readings to
resume.

Rehearsal check: stop the source and verify stale values clear too.

Signal handling is implemented. Its behavior on the assembled wearable must still be verified.

Measure accuracy. Then make the claim.

PAIR

PULSE + reference device
Same moment. Same condition.

REPEAT

Rest, exercise and recovery.
Multiple sessions and participants.

VALIDATION PROTOCOL

Record

Reference model, PULSE HR, time, phase

Calculate

Signed error and mean absolute error

Retain

Session evidence + exported report

No measured accuracy results are claimed here. Physical validation is the next milestone.

From a hackathon winner to a tested wearable.

BUILT

Sensor-to-dashboard pipeline.
Sessions and evidence export.

NEXT

Secure the enclosure.
Run reference-device trials.

SEEKING

Feedback on the protocol.
A supervised pilot opportunity.

See the effort. Understand the recovery.

Research prototype: uncalibrated SpO2 estimate, motion-based respiration and a fatigue model requiring validation.
No clinical or injury-prevention claim.

github.com/006ZERO/PULSE

